# Two boards forget their starts

## Mathematics before the session

A board is a length-three word in R and B. A color-setting instruction (i,c) replaces slot i by c on **both** boards, even if a slot already has color c. For any fixed instruction story, the final color at a visited slot is its last assigned color; an unvisited slot keeps its starting color. If u slots are unvisited, exactly $2^u$ different final boards can arise from the eight starts.

Two boards can be made to agree with exactly as many chosen color-setting instructions as their number of initially disagreeing slots (their Hamming distance). Each differing slot needs an instruction; one instruction at each such slot suffices. Thus the minimum ranges from zero to three. Once a slot agrees, later shared assignments cannot make it disagree.

A fixed story makes **every** pair of starts agree if and only if it visits every slot. The shortest such story has three assignments, one per slot. Two particular starts can agree sooner if every unvisited slot already agreed initially. The every-start guarantee and the matching time of one pair are different statements.

## Toggle contrast and mixed stories

A shared toggle changes R to B and B to R in the named slot on both boards. It preserves the entire set of disagreements. Thus two distinct boards can never merge under shared toggles alone. With assignments and toggles mixed, all starts give the same finish if and only if **each slot receives at least one color-setting instruction**. After that slot’s last assignment, its final color is the assigned color flipped once for each later toggle there. A slot with only toggles still remembers its starting color, possibly complemented.

## Random instructions and limits

For independent uniform draws from the six (slot,color) cards, positions are independent uniform choices and fresh colors are independent fair choices, independent of the positions. Conditional on any fixed slot story visiting all slots, the final colors are the independent colors at the three distinct last-visit times. Thus all eight final boards have probability 1/8, regardless of the start.

No finite number of random draws guarantees that every slot has been visited. Even ten draws can all name the same slot. With the stated independent uniform mechanism, all slots are eventually visited with probability one: after m draws, the chance some slot is missing is at most $3(2/3)^m$, which tends to zero. This is almost-sure eventual forgetting, not a fixed finite deadline. The uniform-color conclusion must not be extended to a stopping rule that chooses when to stop by inspecting colors.

## What each group is discovering

**K-1:** repair disagreements and distinguish chosen instructions from random repetitions. **Grades 2-3:** universal guarantees, visited-slot markers, and the contrast with toggles. **Grades 4-5:** count surviving dependence, prove the mixed-story criterion, and use last assignments to explain an exact uniform distribution.

# Running the session

## Materials and readiness

For eleven children in five pairs (at the Grades 4-5 table a pair and a referee), prepare ten three-slot paper strips, thirty reversible R/B counters, and five cups. Counters must be at most 1.4 cm across to fit the printed working cells, the smallest 1.6 cm (K-1 Problem 4); with standard 2.5 cm counters, play on strips with 3 cm slots. If counters are not reversible, use loose R and B paper squares, with enough of each to overwrite a slot. Each pair needs the six instruction cards 1R, 1B, 2R, 2B, 3R, 3B: cut them from a spare copy of the random-draw page (K-1 p. 4, Grades 2-3 p. 3, Grades 4-5 p. 5) or write them on same-size tickets. Supply three small markers per pair and pencils.

Print the packets: four pages for K-1 and Grades 2-3, five for Grades 4-5. All groups need left-to-right positions and color matching. Middle work adds tracking visited slots and a simple change of rule. Upper work uses all eight board states and independent equal-weight choices; fraction notation is optional. Test a setting instruction and a toggle yourself before the session so their different effects are unmistakable. Prepared, unpiloted material.

## Short shared launch

Let children build two rows, BBR and RBB. Apply “slot 1 becomes B” to both: the first stays BBR, because its slot 1 is already blue, and the second becomes BBB. This pair is not on any page. Before the random page, demonstrate drawing one card, applying it to both boards, returning it, and mixing.

When the toggle rule is introduced, reset RBR and BRB. Toggle slot 2 to get RRR and BBB: both counters change. Put the setting example beside the toggle example. Add a visited marker only when a color-setting story first names that position; the marker is memory of an instruction, not a claim that a visible color changed.

## One board for each child

Each child in a pair owns one board. One child reads every instruction or toggle aloud, both apply it to their own board, and then they compare. At the Grades 4-5 table the third child referees, checking that each instruction went onto both boards. This matters most for toggles: a child who flips only the differing counter can “match” RBR and RRR, which shared toggles never can.

# Timing, hints and source

## About 40 minutes at the tables

**Launch:** about 5 minutes, whole group. **Tables, 0-5 minutes:** pairs set up two boards, one owner each, and try a few chosen instructions. **5-35:** the first route. K-1: Problems 1 and 2 (fewest repairs), Problem 4 (two instructions that sometimes match), then the random draws of Problems 6-7. Grades 2-3: Problems 1-3 (fewest repairs, which stories guarantee a match, matching before every slot is marked). Grades 4-5: Problems 1-3 (the distance rule, counting final boards, when a story forgets). **35-40:** show one guarantee and one counterexample. **Sharing, a few minutes:** “Once two boards match, can a shared instruction make them differ?” Later problems are return visits.

For older groups, aim for one fully justified every-start criterion before attempting probabilities. A handful of trials can reveal repeated slots; an explicit unvisited-slot counterexample proves there is no fixed-draw guarantee.

## Optional hints and source

“Which starting slots are still untouched?” “Can one instruction repair two different slots?” “If we start opposite at a slot never assigned a color, what can toggles do?” “Which instruction was the last one to set this final slot?”

Gautam Iyer, Carnegie Mellon probability notes, Coupling, subsection “Lazy random walk on the binary hypercube,” Proposition 9 proof sketch. The source uses shared coordinate refresh and coordinate coverage. The three-slot setting/toggle comparison and exact keyed examples are independently derived adaptations. Coverage is a sufficient coupling time; a particular pair may couple earlier.

# K to 1 answer key

## Problem 1

RBR versus BRB differs at all three slots, so it needs **three** instructions. For example, set slots 1, 2, 3 to R; both finish RRR. No instruction can change more than one slot, so fewer cannot repair all three differences. No pair needs more: assigning all three slots always works.

## Problem 2

| Starting pair in reading order | Fewest instructions | One possible repair |
|---|---|---|
| RBR / RRR | 1 | 2 becomes R |
| RRB / BRR | 2 | 1 becomes R; 3 becomes R |
| BRB / BRB | 0 | Already equal |
| RRR / BBB | 3 | 1, 2, 3 each become R |

## Problem 3

Example needing exactly two: RRR versus BBR. They disagree in slots 1 and 2; set those slots to R. No pair needs four because there are only three slots.

## Problem 4

The story sets slot 1 to R and slot 3 to B. Two different starts that match afterward: **RRR / BRB**, both finishing RRB. A pair that stays different: **RRR / BBB**, finishing RRB and RBB. Exactly the initial middle color survives, so the starts finish alike precisely when they started with the same middle color.

## Problem 5

Choose one setting instruction per slot, for example 1 becomes R, 2 becomes B, 3 becomes R. Every start ends RBR. Two instructions cannot guarantee this: at least one slot is never set, and two starts differing at that slot remain different there.

## Problem 6

Random results vary. Yes, matching can take more than three draws. From RRR/BBB, drawing “1 becomes R” six times leaves RRR/RBB, still different. A chosen story visiting all three slots would work in three instructions, but a random draw can repeat a previously visited slot. Six draws are exploration, not a guaranteed completion rule.

## Problem 7

No. When the boards match, a shared instruction puts the same color into the same slot of each and leaves all other matching slots alone. Agreement therefore persists after every later shared instruction.

# Grades 2 to 3 answer key

## Problem 1

RBR/BRB needs three instructions, one at each differing slot. Examples needing fewer: RRR/RRR needs zero; RBR/RRR needs one; RRR/BBR needs two. No pair needs more than three, since one setting instruction per slot guarantees agreement. The minimum equals the number of disagreements.

## Problem 2

Only the **second** printed story guarantees every starting pair matches. Its sequence is 2→R, 3→B, 1→R: every slot is assigned, so every start finishes RRB. The first story, 1→R, 1→B, 3→R, leaves slot 2 untouched; the third, 2→B, 2→R, 2→B, leaves slots 1 and 3 untouched.

For counterexamples, start RRR/BBB. Under the first story the finishes are BRR/BBR, which differ at slot 2. Under the third they are RBR/BBB, differing at slots 1 and 3. The slots marked beside the stories (Problem 3) are {1,3}, {1,2,3}, and {2} respectively.

## Problem 3

Yes. RRR/BRR becomes a matching RRR pair after 1→R, even though slots 2 and 3 are unmarked: two different starting boards matching before every slot is marked. Those slots agreed initially. In contrast, RRR/BBB under the same one-instruction story finishes RRR/RBB and does not match. For a fixed pair it is enough to visit all initially differing slots.

## Problem 4

A counterexample is 1→R, 1→R, 1→R from RRR/BBB, leaving RRR/RBB. Ten repeats of 1→R also fail. Any fixed number of draws could miss a slot; replacing tickets makes repeated draws legal. This refutes “must,” even if many actual trials do finish quickly.

## Problem 5

A shortest universal story has **three** instructions, one assignment at each slot; their colors and order are arbitrary. Example: 1→R, 2→B, 3→R. Any shorter story omits a slot. Starts differing there cannot finish alike, so fewer cannot guarantee agreement for every pair.

## Problem 6

Of the three printed pairs, only the **third**, BRB/BRB, can match; it already does and remains matched under every shared toggle story. RBR/RRR differs at slot 2; RRB/BRR differs at slots 1 and 3. Toggling both bits of a disagreement just exchanges R/B with B/R, so those disagreements persist.

## Problem 7

A mixed story that always works: 1→R, toggle 1, 2→B, 3→R. Every start finishes BBR. A mixed story that can fail: 1→R, toggle 2. From RRR/BBB it produces RBR/RRB. Every slot must have at least one color-setting instruction to erase all starting information; shared toggles neither erase a disagreement nor undo an agreement created by a setting instruction.

# Grades 4 to 5 answer key

## Problem 1

The displayed pairs need 1, 2, 0, 3 instructions in reading order. In general the minimum is the number of initially differing slots. Each needs an assignment, and one there suffices.

## Problem 2

| Story | Possible finishes | Untouched slots |
|---|---|---|
| 1→R, 1→B, 3→R | BRR, BBR (2) | 2 |
| 2→R, 3→B, 1→R | RRB (1) | None |
| 2→B, 2→R, 2→B | RBR, RBB, BBR, BBB (4) | 1 and 3 |

Each visited slot is fixed by its last assignment. The unvisited slots can independently retain either starting color, giving $2^u$ finishes for u unvisited slots.

## Problem 3

A color-setting story forgets every start exactly when all three slots are visited. Sufficiency follows from last assignments. Necessity: an untouched slot preserves its starting color, so starts differing there finish differently. Two different boards may agree sooner: RRR and BRR match after 1→R.

## Problem 4

No. Of the printed pairs, the first two keep their disagreements under every toggle story; BRB/BRB already matches. A shared toggle preserves equality or inequality at its slot. Induct over the story: the disagreement set never changes. So distinct starts cannot merge.

## Problem 5

The criterion is at least one color-setting instruction at **each** slot. If a slot is assigned, its last assignment fixes its color independently of the start, and later toggles only flip that known color. If a slot is never assigned, its final color is its initial color or its complement, according to the parity of its toggles; two starts opposite there remain opposite. These prove both directions. Printed stories: the first always finishes BBR; the second finishes RBB or RRB (slot 2 is only toggled, twice); the third always finishes BRR (slot 3 is set to B, then toggled once).

## Problem 6

All eight boards tie, with probability 1/8. For the slot story 1,3,1,2, write the four fresh colors c1,c2,c3,c4. The final board is (c3,c4,c2); c1 is overwritten. Each target board fixes c2,c3,c4, while c1 has two choices. Thus every final board is produced by exactly two of the 16 equally likely color stories, whatever the starting colors.

## Problem 7

Yes, for any fixed slot story visiting every slot: three distinct last-visit times supply independent fair colors. No, four random instructions need not visit every slot. Four draws of 1→R leave slots 2 and 3 unchanged.

# Starting routes and return visits

Planning update of 4 October 2026. The timing menus are examples; a theme can continue over several meetings. Keep the full Grades 4-5 investigations and choose by readiness.

### Use and return choices

Keep one adult at each table to read tickets and record a child-chosen story; each child owns one board. First routes, in about 40 minutes at the tables (page 3): K-1 P1, P2, P4 and P6-7, repairs, a two-instruction story and random draws; grades 2-3 P1-3; grades 4-5 P1-3. On return, K-1 can invent pairs and choose a universal three-instruction story (P3, P5). Grades 2-3 can separate the no-deadline claim (P4-5) from toggles and mixed stories (P6-7). Grades 4-5 can prove the mixed-story criterion with the printed stories (P4-5), then explain the last-assignment color distribution (P6-7). Preserve visited markers as a record of assignments, not a tally of visible changes.